

EC 441 Midterm Review: Solutions

Introduction to Computer Networking — Spring 2026

Boston University

Instructor Note

This document contains complete worked solutions to all review problems. It is **not for distribution to students** before the exam.

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1 Network Fundamentals & Delays

Solution: Problem 1.1 — End-to-End Delay

Given: $L = 5 \text{ MB} = 5 \times 10^6 \times 8 = 4 \times 10^7 \text{ bits}$, $R = 10 \text{ Mb/s} = 10^7 \text{ b/s}$, $d = 1500 \text{ km}$, $v = 2 \times 10^8 \text{ m/s}$, $N = 3 \text{ links}$ (2 store-and-forward routers).

(a) **Transmission delay per link:**

$$d_{trans} = \frac{L}{R} = \frac{4 \times 10^7 \text{ bits}}{10^7 \text{ b/s}} = 4 \text{ s}$$

(b) **Propagation delay per link:**

$$d_{prop} = \frac{d}{v} = \frac{1.5 \times 10^6 \text{ m}}{2 \times 10^8 \text{ m/s}} = 7.5 \text{ ms}$$

(c) **Total end-to-end delay** (store-and-forward across $N = 3$ links):

$$d_{total} = N \cdot d_{trans} + N \cdot d_{prop} = 3 \times 4 \text{ s} + 3 \times 0.0075 \text{ s} = 12 \text{ s} + 0.0225 \text{ s} \approx \mathbf{12.02 \text{ s}}$$

(d) **Dominant delay:** Here, $d_{trans} = 4 \text{ s} \gg d_{prop} = 7.5 \text{ ms}$: **transmission delay dominates**. A 5 MB file over a 10 Mb/s link is slow to push onto the wire.

For the alternative ($L = 100 \text{ bytes}$, $d = 5000 \text{ km}$):

$$d_{trans} = \frac{800 \text{ bits}}{10^7 \text{ b/s}} = 80 \mu\text{s}, \quad d_{prop} = \frac{5 \times 10^6 \text{ m}}{2 \times 10^8 \text{ m/s}} = 25 \text{ ms}$$

Here $d_{prop} = 25 \text{ ms} \gg d_{trans} = 0.08 \text{ ms}$: **propagation delay dominates**. A tiny packet crosses a vast distance—it zips onto the wire instantly, then waits for the signal to travel.

Solution: Problem 1.2 — Protocol Layers

(a) From top to bottom: Application, Transport, Network, Link, Physical.

(b)

- (i) Breaking into segments: **Transport** layer (TCP segmentation).
- (ii) Adding IP addresses: **Network** layer (IP datagram).
- (iii) Determining next-hop MAC: **Link** layer (ARP, then Ethernet framing).

(c) As data passes *down* the stack, each layer **encapsulates** the data unit from the layer above by prepending its own header (and sometimes appending a trailer). The payload from layer $N + 1$ becomes the data field of layer N 's protocol data unit. At the receiver, each layer strips its header as the data moves back up the stack.

2 Information Theory & Entropy

Solution: Problem 2.1 — Shannon Entropy

(a) $p_A = 1/2$, $p_B = 1/4$, $p_C = 1/8$, $p_D = 1/8$.

$$\begin{aligned} H(X) &= - \sum_i p_i \log_2 p_i \\ &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} \\ &= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) \\ &= 0.5 + 0.5 + 0.375 + 0.375 = \mathbf{1.75 \text{ bits/symbol}} \end{aligned}$$

(b) Maximum entropy for $M = 4$ symbols:

$$H_{max} = \log_2 4 = \mathbf{2 \text{ bits/symbol}}$$

achieved when all four symbols are **equally probable**: $p_i = 1/4$ for all i . The distribution in (a) is slightly non-uniform, so $H = 1.75 < 2$.

(c) Entropy is the *minimum average number of bits* per symbol achievable by any lossless compression scheme (equivalently, the information content per symbol of the source).

Solution: Problem 2.2 — Multimedia Data Rates

(a) CD audio:

$$R = 44,100 \text{ Hz} \times 16 \text{ bits/sample} \times 2 \text{ channels} = 1,411,200 \text{ b/s} \approx \mathbf{1.41 \text{ Mb/s}}$$

(b) Compression ratio:

$$\text{ratio} = \frac{1.5 \text{ Gb/s}}{8 \text{ Mb/s}} = \frac{1500}{8} \approx \mathbf{188 : 1}$$

This is typical for H.264 at streaming quality; practical ratios range from 100:1 to 300:1.

(c) A US average home internet downstream speed is roughly 100–500 Mb/s; most connections are 25–100 Mb/s. Streaming 1.5 Gb/s uncompressed would require bandwidth 15–60× beyond what most home connections provide. Storage cost would also be prohibitive (~67 GB/hour). Compression makes streaming feasible.

Solution: Problem 2.3 — Units and Prefixes

(a) $2 \text{ TB} = 2 \times 10^{12} \text{ bytes} = \mathbf{2,000,000,000,000}$ bytes (SI/decimal).

(b) $1 \text{ TiB} = 2^{40} \text{ bytes} \approx 1.0995 \times 10^{12} \text{ bytes}$.

$$1.82 \text{ TiB} = 1.82 \times 1.0995 \times 10^{12} \approx 2.001 \times 10^{12} \text{ bytes}$$

Yes, consistent. The manufacturer reports in decimal (SI) terabytes; the OS reports in binary (IEC) tebibytes. The same physical capacity reads as a slightly smaller number in TiB than in TB.

(c) $1 \text{ Gb/s} = 10^9 \text{ b/s}$. $1 \text{ GiB} = 2^{30} \times 8 \text{ bits} = 8,589,934,592 \text{ bits}$.

$$t = \frac{8,589,934,592 \text{ bits}}{10^9 \text{ b/s}} \approx \mathbf{8.59 \text{ s}}$$

Note: $1 \text{ GiB} \neq 1 \text{ Gb}$; be careful with decimal vs. binary prefix and bits vs. bytes.

3 Physical Layer: Guided Media

Solution: Problem 3.1 — Attenuation and dB

(a) Total attenuation:

$$A = 0.3 \text{ dB/km} \times 5 \text{ km} = \mathbf{1.5 \text{ dB}}$$

(b) $P_{in} = 10 \text{ mW}$.

$$P_{out}[\text{dBm}] = P_{in}[\text{dBm}] - A = 10 \text{ dBm} - 1.5 \text{ dB} = 8.5 \text{ dBm}$$

$$P_{out} = 10^{8.5/10} \text{ mW} = 10^{0.85} \approx \mathbf{7.08 \text{ mW}}$$

Alternatively: $P_{out}/P_{in} = 10^{-1.5/10} = 10^{-0.15} \approx 0.708$, so $P_{out} = 10 \times 0.708 = 7.08 \text{ mW}$.

(c) $P_{in} = 10 \text{ mW}$:

$$P_{in}[\text{dBm}] = 10 \log_{10} \left(\frac{10 \text{ mW}}{1 \text{ mW}} \right) = 10 \log_{10}(10) = \mathbf{10 \text{ dBm}}$$

Solution: Problem 3.2 — Shannon-Hartley Theorem

(a) SNR linear:

$$\text{SNR} = 10^{40/10} = 10^4 = \mathbf{10,000}$$

(b) Shannon capacity:

$$C = B \log_2(1 + \text{SNR}) = 10^6 \times \log_2(10,001) = 10^6 \times \frac{\ln(10,001)}{\ln 2} \approx 10^6 \times 13.29 \approx \mathbf{13.3 \text{ Mb/s}}$$

(c) $20 \text{ Mb/s} > 13.3 \text{ Mb/s} = C$. This is **physically impossible**. Shannon's theorem is an absolute upper bound; no coding or modulation scheme can achieve reliable communication above capacity.

Solution: Problem 3.3 — Digital Signaling**(a)**

- Unipolar NRZ: 0 V for '0', $+V$ for '1'.
- Polar NRZ-L: $-V$ for '0', $+V$ for '1'.

Polar NRZ is preferred because it is *balanced* (zero DC component), allowing AC coupling. It also offers better noise immunity: the symbol separation is $2V$ vs. V for unipolar.

(b) Manchester encoding:

- '1': **low-to-high transition** at the mid-bit boundary.
- '0': **high-to-low transition** at the mid-bit boundary.

(c)

- Binary polar NRZ at 100 Mb/s: 1 bit/symbol $\Rightarrow$ symbol rate = **100** Msymbols/s.
- 16-PAM at same symbol rate: $\log_2(16) = 4$ bits/symbol $\Rightarrow$ bit rate = $100 \times 4 =$ **400** Mb/s.

(d)

- **Advantage:** $4\times$ higher bit rate for the same channel bandwidth (spectral efficiency).
- **Disadvantage:** 16 amplitude levels are much closer together than 2, requiring higher SNR to reliably distinguish symbols. Far more susceptible to noise and interference.

4 Wireless Physical Layer

Solution: Problem 4.1 — Free-Space Path Loss and Link Budget

(a) $f = 2400$ MHz, $d = 50$ m = 0.050 km:

$$\begin{aligned} L_{FSPL} &= 32.45 + 20 \log_{10}(2400) + 20 \log_{10}(0.050) \\ &= 32.45 + 20(3.380) + 20(-1.301) \\ &= 32.45 + 67.60 - 26.02 \\ &= \mathbf{74.0} \text{ dB} \end{aligned}$$

(b) Link budget:

$$P_r = P_t + G_t + G_r - L_{FSPL} = 20 + 2 + 0 - 74.0 = \mathbf{-52} \text{ dBm}$$

(c) Link margin = $P_r - P_{sens} = -52 - (-80) = \mathbf{+28}$ dB. Positive margin: **link is viable** with 28 dB of headroom (accounts for walls, multipath, body blockage, etc.).

(d) Path loss increases as $20 \log_{10}(d_2/d_1)$:

$$\Delta L = 20 \log_{10}\left(\frac{200}{50}\right) = 20 \log_{10}(4) = 20 \times 0.602 \approx \mathbf{12} \text{ dB}$$

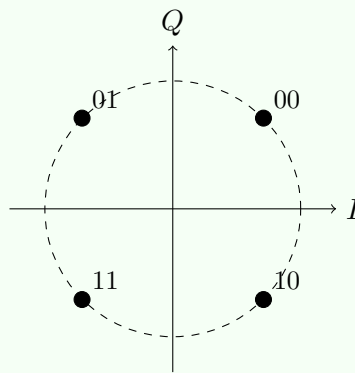
The received power drops by 12 dB (factor of ~ 16 in power), from -52 to -64 dBm.

Solution: Problem 4.2 — Modulation and QAM**(a)**

- QPSK: $\log_2 4 = 2$ bits/symbol.
- 16-QAM: $\log_2 16 = 4$ bits/symbol.
- 64-QAM: $\log_2 64 = 6$ bits/symbol.

(b) $10 \text{ Msymbols/s} \times 6 \text{ bits/symbol} = 60 \text{ Mb/s}$.

(c) Far from the AP, the received SNR is low due to path loss. 256-QAM packs 8 bits/symbol by using 256 tightly spaced constellation points; the minimum required SNR is $\approx 28 \text{ dB}$. At low SNR the receiver cannot reliably distinguish nearby points, causing high BER. BPSK uses only 2 points with maximum separation, requiring as little as 6 dB SNR for acceptable BER. WiFi adaptively selects the modulation and coding scheme (MCS) that maximises throughput while maintaining acceptable error rates.

(d) QPSK constellation (I–Q plane):

Four equally spaced symbols at $(\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$ (or equivalently at $45^\circ, 135^\circ, 225^\circ, 315^\circ$). Gray coding (shown) assigns adjacent symbols bit patterns differing by only one bit.

Solution: Problem 4.3 — Cellular Frequency Reuse

(a) Reuse distance:

$$D = R\sqrt{3N} = 500 \times \sqrt{3 \times 7} = 500 \times \sqrt{21} \approx 500 \times 4.58 \approx \mathbf{2290 \text{ m}}$$

(b) C/I ratio:

$$\frac{C}{I} \approx \frac{(3N)^{n/2}}{6} = \frac{(21)^{3.5/2}}{6} = \frac{21^{1.75}}{6}$$

$$21^{1.75} = e^{1.75 \ln 21} = e^{1.75 \times 3.045} = e^{5.329} \approx 206.$$

$$\frac{C}{I} \approx \frac{206}{6} \approx 34.4 \approx \mathbf{15.4 \text{ dB}}$$

(c) Switching from $N = 7$ to $N = 3$:

- (i) **Capacity per unit area increases:** each cell now uses 1/3 of the total spectrum (vs. 1/7), more than doubling the bandwidth per cell. Cells are also smaller (denser reuse), adding spatial capacity.
- (ii) **Interference increases:** co-channel cells are closer together (D shrinks), so C/I drops. Managing this requires advanced techniques (power control, MIMO, beamforming). Modern LTE/5G networks often operate at $N = 1$ using such techniques.

5 Error Control Coding

Solution: Problem 5.1 — Hamming Distance and d_{min} Theorem

$\mathcal{C} = \{000000, 001110, 110100, 111010\}$

(a) Pairwise Hamming distances:

Pair	Differing positions	d_H
000000 vs. 001110	2, 3, 4	3
000000 vs. 110100	0, 1, 3	3
000000 vs. 111010	0, 1, 2, 4	4
001110 vs. 110100	0, 1, 2, 4	4
001110 vs. 111010	0, 1, 3	3
110100 vs. 111010	2, 3, 4	3

(b) $d_{min} = 3$ (from the table above).

(c) **Detect only:** $e_d \leq d_{min} - 1 = 2$. Can detect up to **2 errors**.

(d) **Correct only:** $e_c \leq \lfloor (d_{min} - 1)/2 \rfloor = \lfloor 1 \rfloor = 1$. Can correct **1 error**.

(e) Check $e_c = 1$, $e_d = 2$ simultaneously:

d_{min} theorem requires $e_c + e_d \leq d_{min} - 1$: $1 + 2 = 3 \leq 2$? **No**. The theorem is violated. This code *cannot* simultaneously detect 2 errors and correct 1. (With $d_{min} = 3$: can correct 1 *or* detect 2, but not both at the same time.)

Solution: Problem 5.2 — Code Efficiency Comparison

(a) Probability of ≥ 1 error in 1000 bits:

$$P_{frame} = 1 - (1 - p)^{1000} \approx 1 - e^{-1000 \times 10^{-5}} = 1 - e^{-0.01} \approx \mathbf{0.00995 \approx 1\%}$$

(b) Parity code: code rate $R_c = 1000/1001 \approx \mathbf{0.999}$.

Undetected error requires an *even* number of errors. Dominant term: exactly 2 errors.

$$P_{undetected} \approx \binom{1001}{2} p^2 \approx \frac{1001 \times 1000}{2} \times (10^{-5})^2 \approx 5.005 \times 10^5 \times 10^{-10} \approx \mathbf{5 \times 10^{-5}}$$

(c) Hamming code: $n = 1010$, $k = 1000$. Code rate $R_c = 1000/1010 \approx \mathbf{0.990}$.

This is lower than the parity code's rate of 0.999 — nearly ten times more overhead.

(d) **Parity/ARQ (Scheme A) provides higher throughput.** At $p = 10^{-6}$, frame errors are rare ($\sim 1\%$). ARQ retransmission overhead is minimal. Scheme A wastes only 1 bit per 1000 on redundancy ($R_c = 0.999$), whereas Scheme B wastes 10 bits ($R_c = 0.990$). The error performance of both schemes is similar ($P_e \approx 10^{-9}$). This is the classic result: detection + ARQ is $\approx 5\times$ more efficient than FEC on low-BER bidirectional links.

Solution: Problem 5.3 — CRC by Hand

$M = 1101$, $G(x) = x^3 + x + 1 \leftrightarrow 1011$, $r = 3$.

(a) Append 3 zeros: $x^3 M(x) \leftrightarrow 1101000$

(b) Modulo-2 long division of $1101000 \div 1011$:

$$\begin{array}{r}
 1011 \overline{) 1101000} \\
 \underline{1011} \\
 1100 \\
 \underline{1011} \\
 1110 \\
 \underline{1011} \\
 1010 \\
 \underline{1011} \\
 001
 \end{array}$$

Remainder $R(x) = 001$.

Step-by-step detail:

1. $1101 \oplus 1011 = 0110$; bring down 0 $\rightarrow 1100$
2. $1100 \oplus 1011 = 0111$; bring down 0 $\rightarrow 1110$
3. $1110 \oplus 1011 = 0101$; bring down 0 $\rightarrow 1010$
4. $1010 \oplus 1011 = 0001$; no more bits. Remainder = 001.

(c) Transmitted frame:

$$T = x^3 M(x) + R(x) \leftrightarrow 1101000 \oplus 000001 = \mathbf{1101001}$$

(d) Flip bit 3 (0-indexed, leftmost = 0): $1101001 \rightarrow 1100001$.

Divide $1100001 \div 1011$:

1. $1100 \oplus 1011 = 0111$; bring down 0 $\rightarrow 1110$
2. $1110 \oplus 1011 = 0101$; bring down 0 $\rightarrow 1010$
3. $1010 \oplus 1011 = 0001$; bring down 1 $\rightarrow 0011$
4. Leading bit = 0; no XOR. Remainder = $011 \neq 000$.

Remainder $\neq 0 \Rightarrow$ **error detected.** ✓

6 Multiple Access Protocols

Solution: Problem 6.1 — ALOHA Throughput

(a) With $N = 10$ nodes, each transmitting with probability p :

$$P(\text{success}) = \binom{N}{1} p(1-p)^{N-1} = 10p(1-p)^9$$

(b) Using the continuous approximation $S = Ge^{-G}$ with $G = Np$:

$$\frac{dS}{dG} = e^{-G}(1-G) = 0 \Rightarrow G = 1 \Rightarrow p = \frac{G}{N} = \frac{1}{10} = \mathbf{0.1}$$

(c) Maximum throughput:

$$S_{\max} = 1 \cdot e^{-1} = \frac{1}{e} \approx \mathbf{0.368} = \mathbf{36.8\%}$$

(d) Pure ALOHA maximum: $S_{\max} = 1/(2e) \approx 0.184 = 18.4\%$.

Slotted ALOHA achieves exactly **twice** the throughput of pure ALOHA because the collision window is halved (one slot T vs. two slot-times $2T$ for pure ALOHA).

Solution: Problem 6.2 — CSMA/CD Minimum Frame Size

(a) One-way propagation delay:

$$\tau = \frac{d}{v} = \frac{2500 \text{ m}}{2 \times 10^8 \text{ m/s}} = \mathbf{12.5 \mu\text{s}}$$

(b) Round-trip:

$$2\tau = \mathbf{25 \mu\text{s}}$$

(c) Minimum frame size from formula:

$$L_{\min} = 2\tau \times R = 25 \times 10^{-6} \text{ s} \times 10 \times 10^6 \text{ b/s} = \mathbf{250 \text{ bits}}$$

(d) $250 \text{ bits}/8 = 31.25 \text{ bytes} \approx 32 \text{ bytes}$.

The IEEE 802.3 standard specifies a minimum of **512 bits (64 bytes)**. The larger standard value accounts for additional delays: up to four repeaters in the path (each adding ~ 4 bit-times), transceiver cable delays, and a safety margin. The formula gives the theoretical minimum; the standard adds a conservative buffer.

(e) Modern Ethernet uses **switches**, which create dedicated, full-duplex, point-to-point links between each device and the switch port. With no shared medium, two devices can never transmit simultaneously on the same segment, so collisions *cannot occur*. CSMA/CD is defined in the standard but never triggered in practice.

Solution: Problem 6.3 — Binary Exponential Backoff

(a) After $k = 1$ collision, backoff range = $[0, 2^1 - 1] = \{0, 1\}$. Each station chooses **0 or 1** independently.

(b) Both stations choose their backoff values uniformly at random from $\{0, 1\}$:

$$P(\text{collision again}) = P(\text{both choose 0}) + P(\text{both choose 1}) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$$

(c) After $k = 2$ collisions, range = $[0, 2^2 - 1] = \{0, 1, 2, 3\}$. Four choices.

(d) CSMA/CD gives up after **16 consecutive collisions**. The frame is **discarded** and the NIC reports a transmission error to the OS/driver. Higher-layer protocols (e.g., TCP) may then retransmit the data. In practice, 16 collisions are essentially impossible on a well-configured network; they would indicate a severe hardware fault.

7 Ethernet (IEEE 802.3)

Solution: Problem 7.1 — Ethernet Frame Analysis

(a) **Broadcast.** Destination MAC is `FF:FF:FF:FF:FF:FF`, the reserved all-ones broadcast address. The least-significant bit (I/G bit) of the first byte is 1, confirming this is a group (not unicast) address. All-ones is specifically the broadcast address (all hosts on the LAN receive the frame).

(b) EtherType `0x0806` = **ARP** (Address Resolution Protocol). This is consistent with the broadcast destination: ARP requests are always broadcast.

(c) Frame size (excluding preamble and SFD):

$$6 + 6 + 2 + 46 + 4 = \mathbf{64 \text{ bytes}}$$
 (minimum Ethernet frame)

(d) The **preamble** (7 bytes of `10101010...`) creates a 10 MHz square wave that allows the receiver's phase-locked loop (PLL) to synchronize its clock to the transmitter's bit rate before the frame arrives. The **SFD** (`10101011`) signals the *end* of preamble and the *start* of the frame; its two consecutive '1' bits are unique within the preamble pattern and are unambiguous.

(e) The FCS (CRC-32) covers: **Destination MAC, Source MAC, Type/Length, and Payload**. It does *not* cover the preamble or SFD (those are stripped before CRC computation).

Solution: Problem 7.2 — ARP Trace

(a) ARP request from A:

- Ethernet source: `AA:AA:AA:AA:AA:AA`; destination: `FF:FF:FF:FF:FF:FF` (broadcast).
- ARP payload: "I am `10.0.0.1`/`AA:AA:AA:AA:AA:AA`. Who has `10.0.0.2`?"
- Broadcast is required because A does not yet know B's MAC address.

(b) ARP reply from B: **unicast** back to A.

- Ethernet source: `BB:BB:BB:BB:BB:BB`; destination: `AA:AA:AA:AA:AA:AA`.
- ARP payload: "`10.0.0.2` is at `BB:BB:BB:BB:BB:BB`."

B learned A's MAC from the request (the ARP request includes the sender's IP and MAC), so the reply can be unicast directly.

(c) Ethernet frame from A to B carrying the IP datagram:

Field	Value
Source MAC	<code>AA:AA:AA:AA:AA:AA</code>
Destination MAC	<code>BB:BB:BB:BB:BB:BB</code>
EtherType	<code>0x0800</code> (IPv4)

(d) Host A uses the **default gateway's MAC** (`GG:GG:GG:GG:GG:GG`) as the Ethernet destination. The IP destination in the packet header is still `192.168.5.1` (C's address), but the Ethernet frame is addressed to the router, which will forward the packet toward C. ARP operates only within a subnet; A cannot ARP for an address on a different subnet.

Solution: Problem 7.3 — Switch Self-Learning

(a) A sends to B; switch table empty.

- S1 **learns**: records $A \rightarrow$ port 1 (source MAC on ingress port).
- S1 **floods**: B is unknown, so S1 sends the frame out ports 2 and 3 (all except ingress port 1).
- Table: $\{A \rightarrow \text{port 1}\}$.

(b) B sends to A; table has $\{A \rightarrow \text{port 1}\}$.

- S1 **learns**: records $B \rightarrow$ port 2.
- S1 **forwards**: A is known on port 1, so frame goes out **port 1 only**.
- Table: $\{A \rightarrow \text{port 1}, B \rightarrow \text{port 2}\}$.

(c) A sends to C; C is not in table.

- S1 refreshes $A \rightarrow$ port 1.
- C is unknown $\Rightarrow$ S1 **floods** out ports 2 and 3.

(d) A loop causes a broadcast frame to travel in circles indefinitely, each switch forwarding it on all ports, which triggers further copies — a **broadcast storm** that consumes 100% of bandwidth and crashes the network. The solution is **Spanning Tree Protocol (STP, IEEE 802.1D)**: switches exchange Bridge Protocol Data Units (BPDUs) to elect a root bridge and logically block redundant links, ensuring the active topology is a tree (loop-free). Blocked ports re-activate automatically if an active link fails.

Solution: Problem 7.4 — Ethernet Generations**(a)**

Standard	Line code
10BASE-T	Manchester encoding
100BASE-TX	4B5B + MLT-3
1000BASE-T	4D-PAM5

(b) “BASE” signifies **baseband** transmission: the signal occupies the full channel bandwidth without frequency-division multiplexing or radio-frequency upconversion. (Contrast with “BROAD” for broadband, used in early cable-TV based Ethernet.)

(c) 100BASE-TX achieves 100 Mb/s over 2 pairs at 125 Msymbols/s using 4B5B+MLT-3 (effective 100 Mb/s after encoding overhead). To reach 1000 Mb/s requires roughly $10\times$ the capacity. 4D-PAM5 uses 5 amplitude levels ($\log_2 5 \approx 2.32$ bits/symbol), but practically delivers ~ 2 data bits/symbol per pair (one level is reserved for trellis-coded FEC). At 125 Msymbols/s per pair:

$$4 \text{ pairs} \times 2 \text{ bits/symbol} \times 125 \text{ Msymbols/s} = 1000 \text{ Mb/s}$$

Thus 4 pairs are necessary because no single-pair encoding can reach 1 Gb/s at 125 Msymbols/s with category 5e cable. Transmitting over all 4 pairs simultaneously achieves the target without exceeding the cable’s bandwidth limitations.